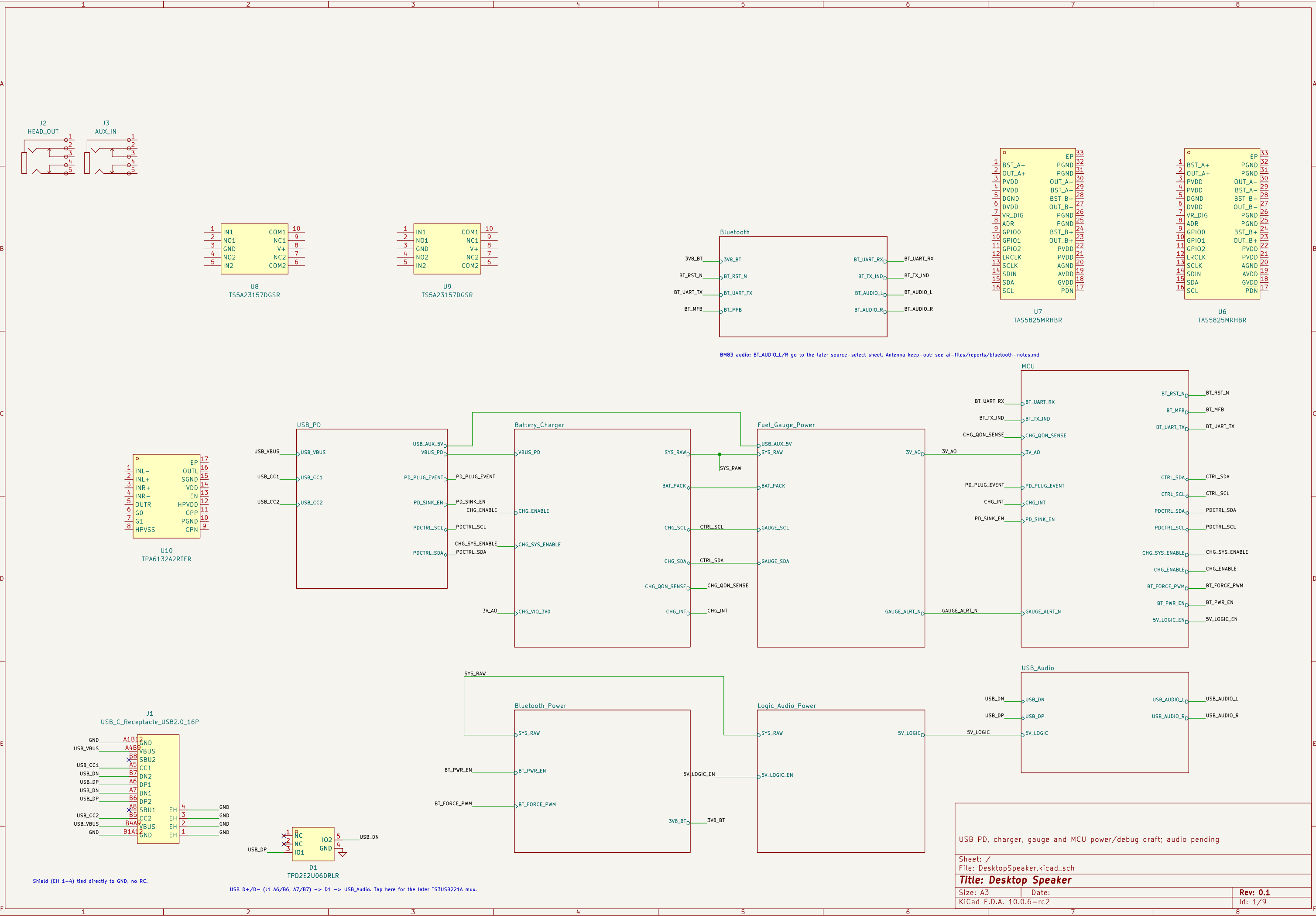
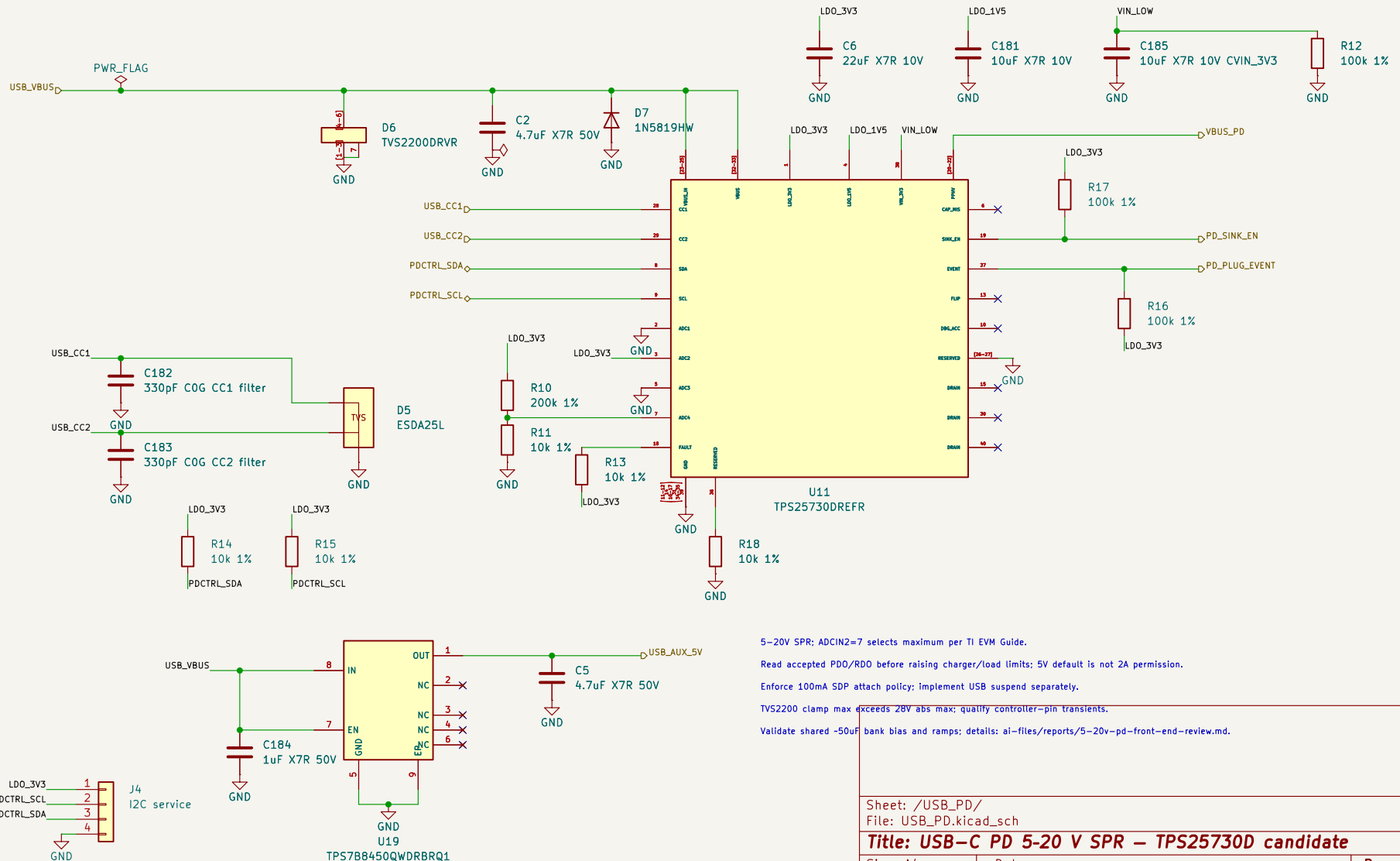
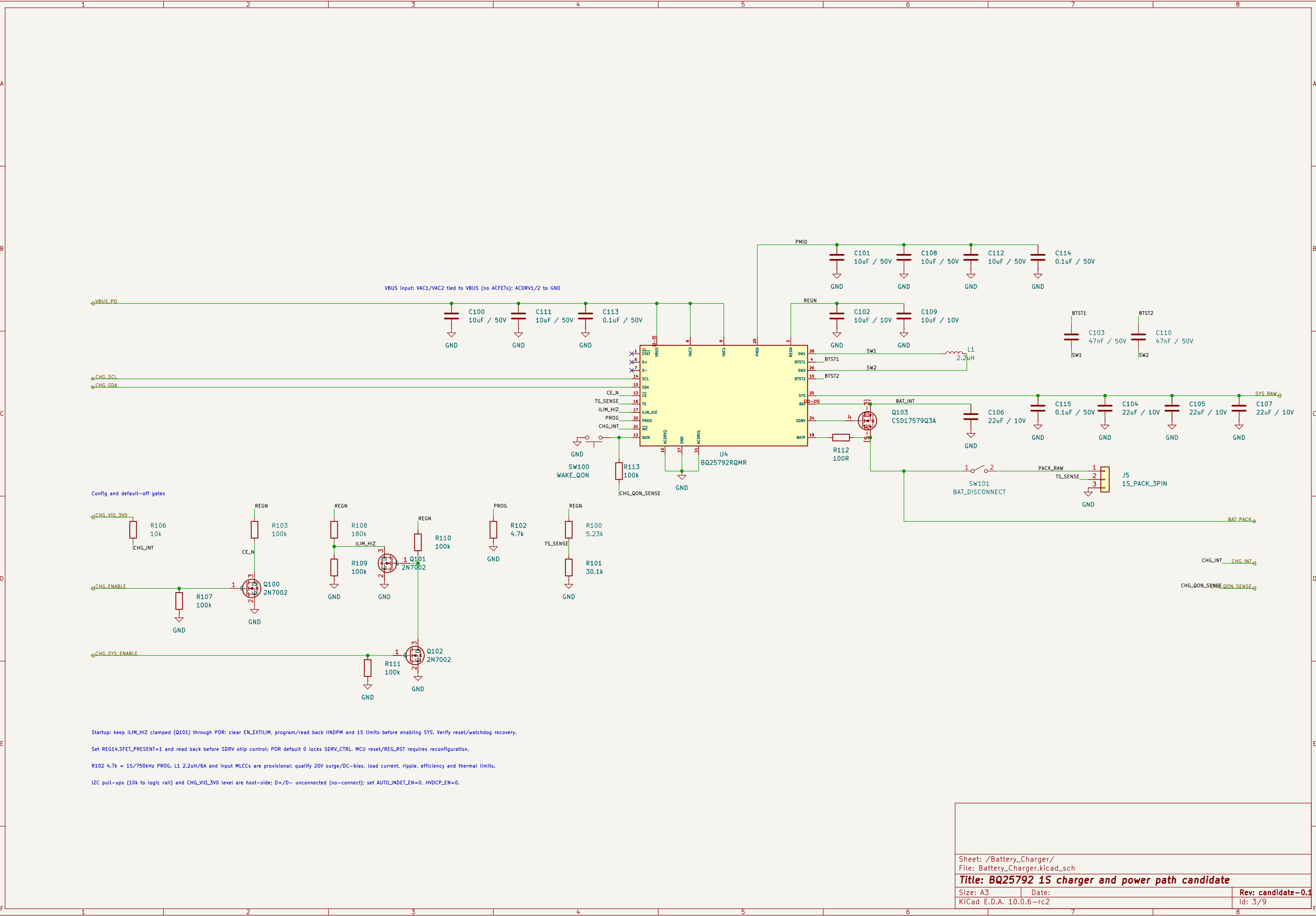
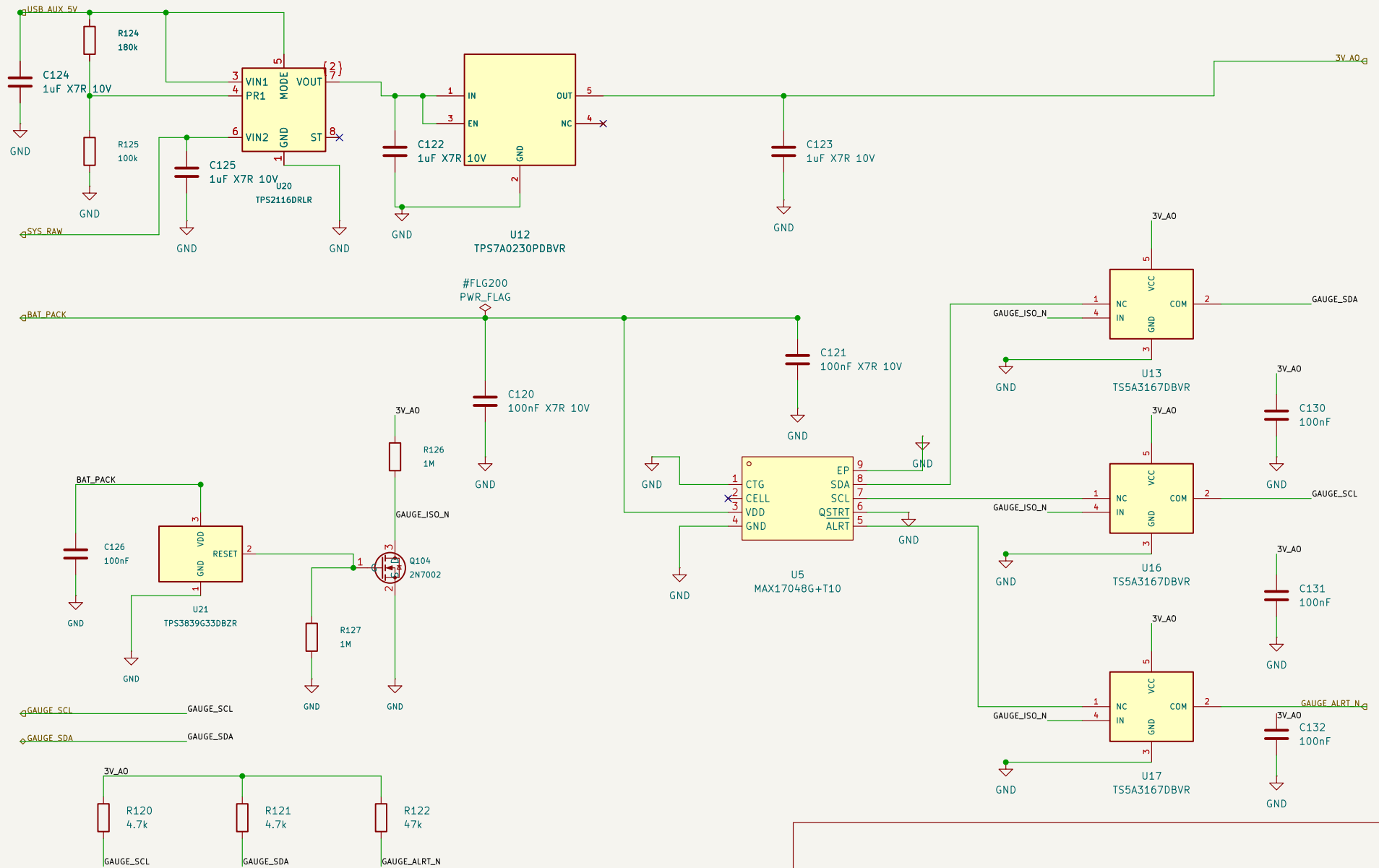










USB—priority logic supply; PR1 nominal 2.8 V.
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus: AO—powered switches; default OFF below valid battery.
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /FuelGauge_Power/
File: FuelGauge_Power.kicad_sch

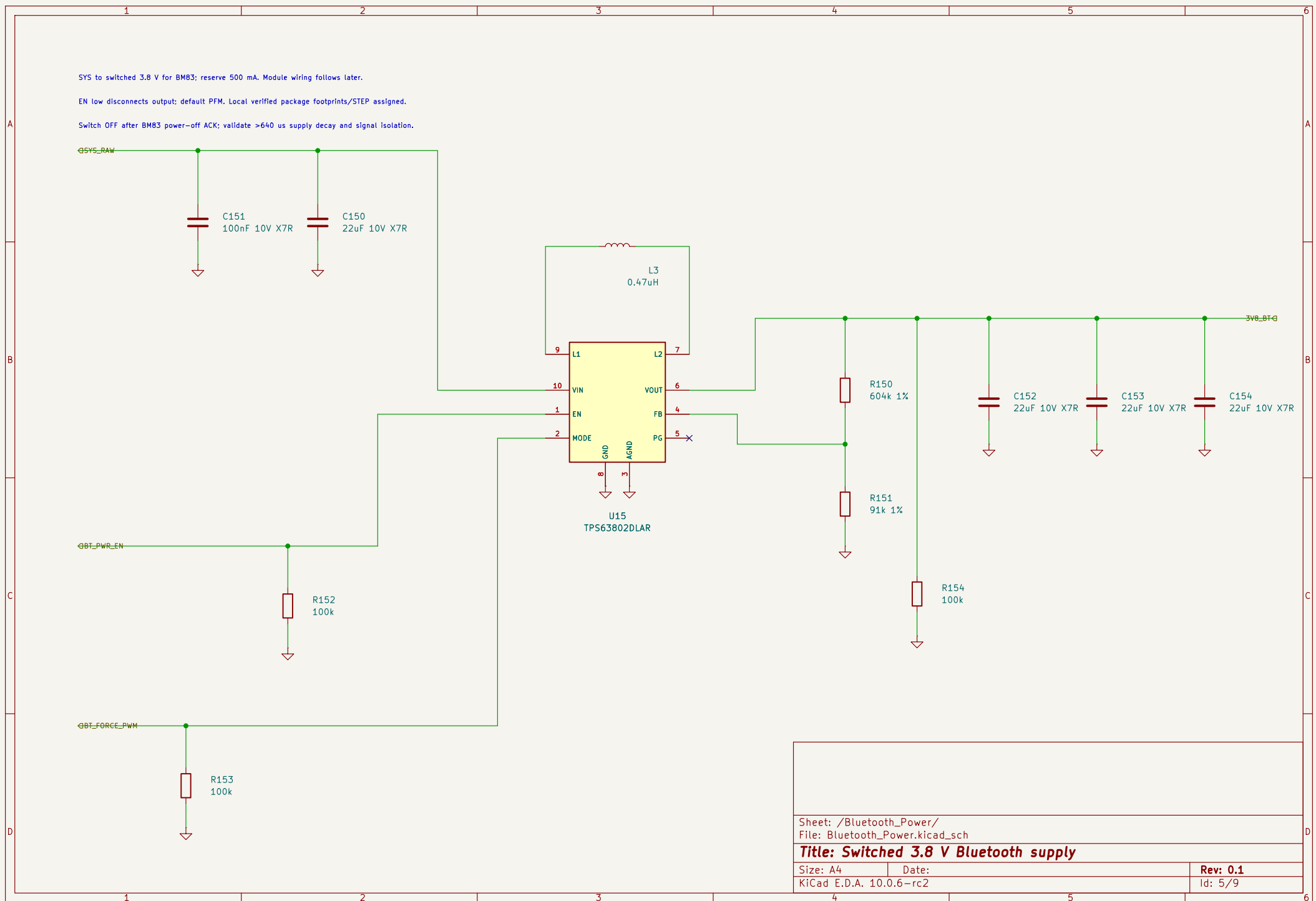
Title: Fuel gauge and low-current power

Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

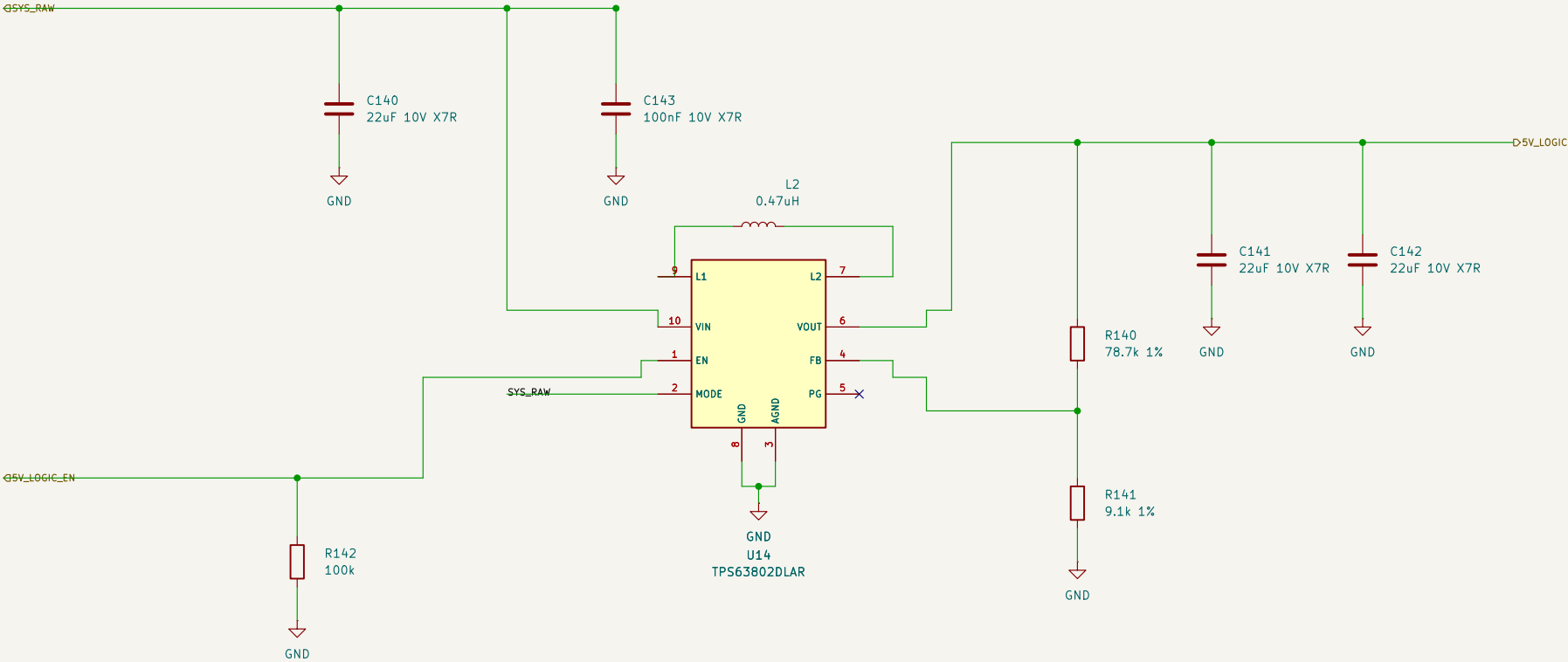
Id: 4/9



Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

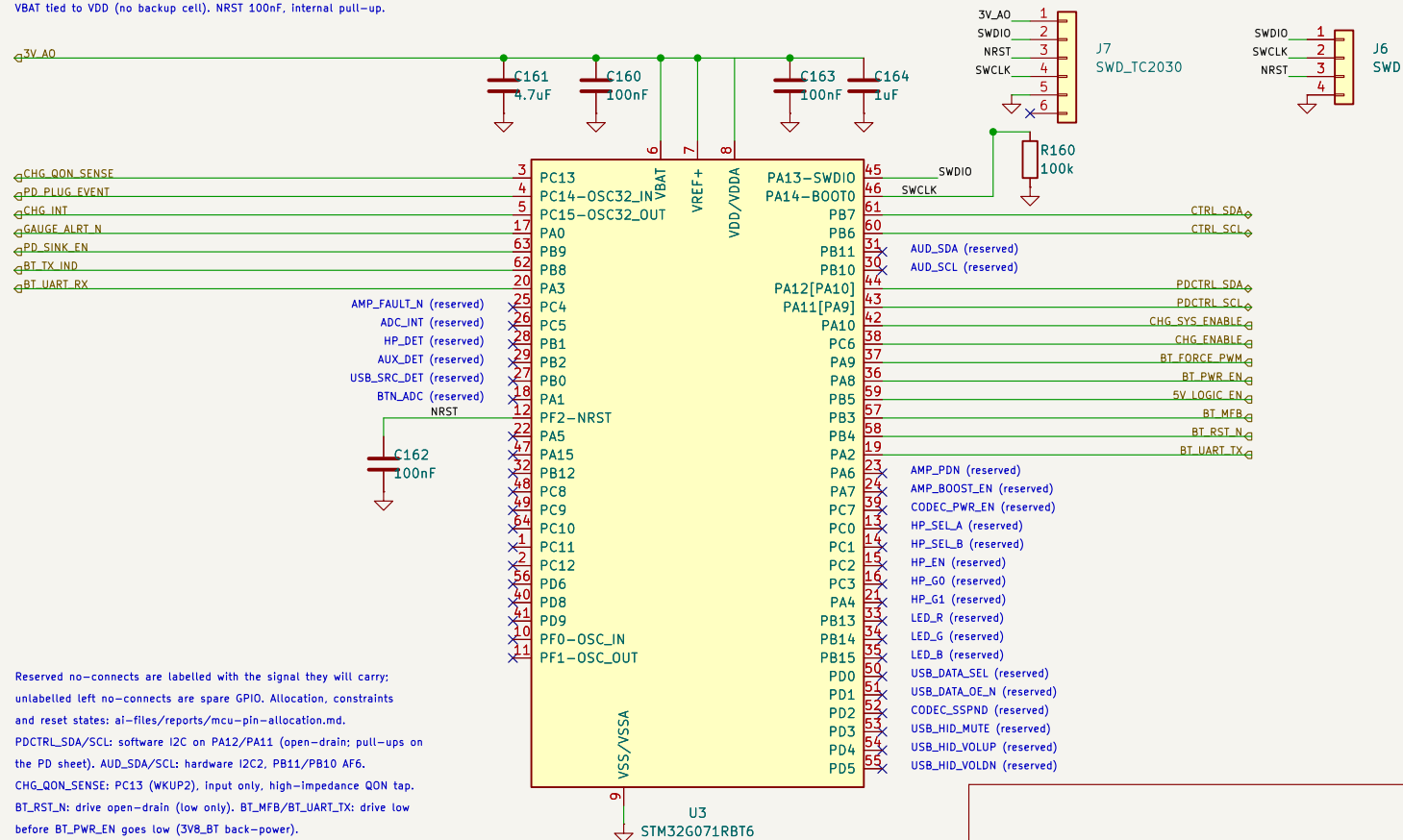
EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_AO supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ 100nF + 1uF (VREFBUF off, VREF+ = VDDA).
VBAT tied to VDD (no backup cell), NRST 100nF, internal pull-up.

SWD: J7 Tag-Connect TC2030-IDC (with legs: 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)



Reserved no-connects are labelled with the signal they will carry;
unlabelled left no-connects are spare GPIO. Allocation, constraints
and reset states: ai-files/reports/mcu-pin-allocation.md.
PDCTRL_SDA/SCL: software I2C on PA12/PA11 (open-drain; pull-ups on
the PD sheet). AUD_SDA/SCL: hardware I2C2, PB11/PB10 AF6.
CHG_QON_SENSE: PC13 (WKUP2), input only, high-impedance QON tap.
BT_RST_N: drive open-drain (low only). BT_MFB/BT_UART_TX: drive low
before BT_PWR_EN goes low (3V8_BT back-power).
BT_TX_JND = module PO_0 (EXTI/wake).

STM32G071RBT6 LQFP64: supply, reset, SWD, power-block links

Sheet: /MCU/
File: MCU.kicad_sch

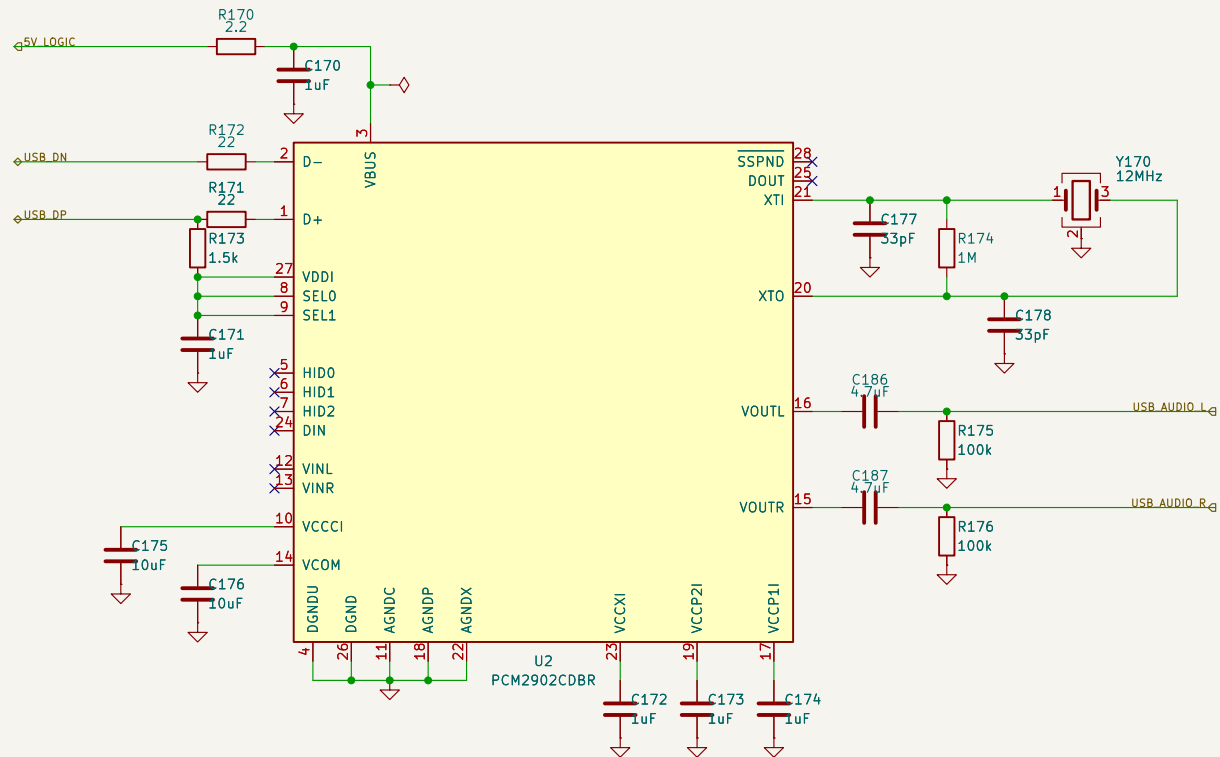
Title: MCU and programming

Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

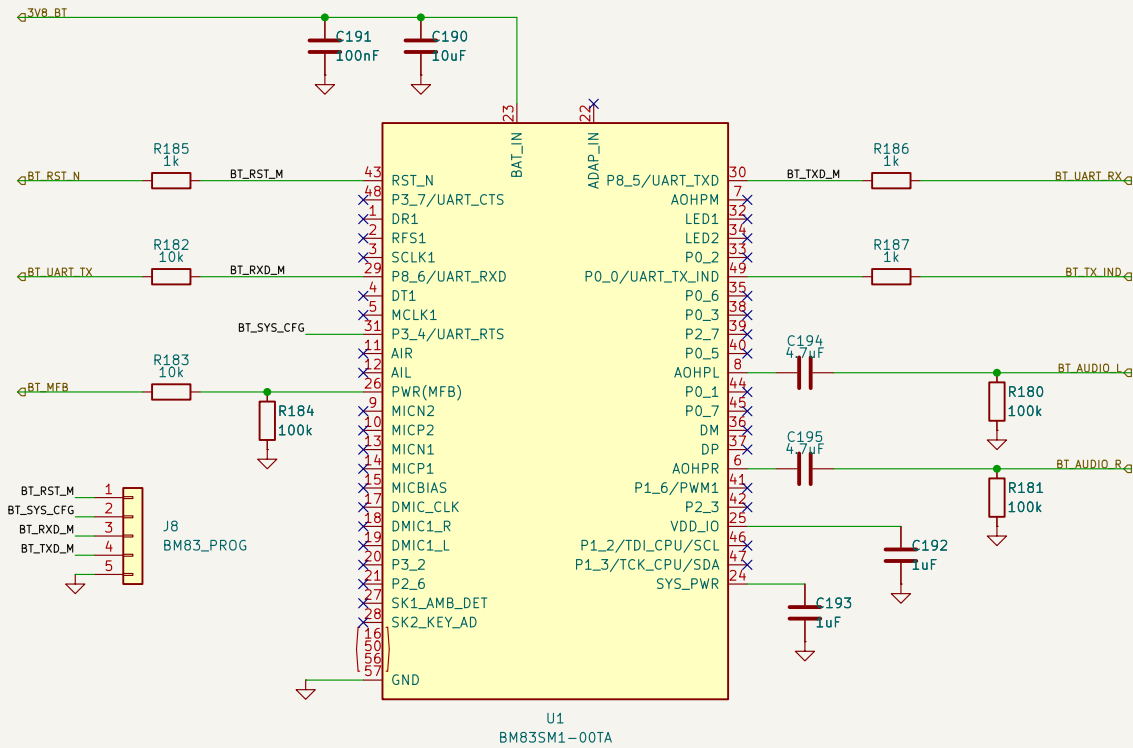
Id: 7/9



PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUD0_L/R

Title: *USB audio*

Rev: 0.1
Id: 8/9



BM83SM1-00TA (DS70005402D). BAT_IN 3.2-4.2 V from 3V8_BT (U15); SYS_PWR/VDD_IO are outputs (decoupling only, do not load or feed). ADAP_IN unused (Internal charger/USB DFU not used).
Host mode UART from the MCU: PB_6 RXD / PB_5 TXD, P0_0 = UART_TX_IND (wakes the MCU). MFB (PWR) = power-on/wake key. RST_N = reset (active low, internal pull-up). Config Tool must set Host mode, P0_0 as TX_IND.
Back-power: MCU lines pass 10k/1k series resistors; firmware must drive PA2/PB3 low (PB4 open-drain/hi-Z) and disable BT_PWR_EN only after the module power-off ACK. Module outputs are 3.0-3.6 V into 5 V tolerant MCU pins.
Audio: single-ended DAC out (AOHPL/AOHPM) AC-coupled, 4.7 uF + 100k bleed, 0.74 Vrms full scale; AOHPM (capless sense) unused. Ports feed the later source-select sheet.
Antenna: integrated PCB antenna, no external RF parts. Keep the antenna end of the module off the board (no copper/components/keep-out) and metal 15 mm clear; see ai-files/reports/bluetooth-notes.md.
J8 programming/test (isUpdate/Config Tool): 1 RST_N, 2 P3_4 SYS_CFG (low at reset = test mode), 3 UART_RXD (adapter TX), 4 UART_TXD (adapter RX), 5 GND. 3V8_BT must be enabled first.

BM83 module: host UART, MFB/RST_N, analogue out

Sheet: /Bluetooth/
File: Bluetooth.kicad_sch

Title: Bluetooth audio

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 9/9